Question	Answer	Marks	Partial Marks
2(a)(iii)	276	**2**	**M1** for $240 \times \left(1+\frac{15}{100}\right)$ oe or **B1** for 36 seen
2(a)(iv)	150	**3**	**M2** for $\frac{240}{100+60}$ [× 100] oe or **M1** for evidence of 160[%] associated 240
2(b)	464 000	**3**	**M1** for $256\,000 \times \left(1+\frac{2}{100}\right)^{30}$ oe **A1** for 463 700 to 463 710 **B1** for *their* more accurate answer seen and rounded to nearest 1000
2(c)	4.5[0]	**3**	**M2** for [x =] $\sqrt[32]{4.09}$ oe or **M1** for $(x)^{32} = 4.09$ oe If 0 scored, **SC2** for answer 3.6 or 3.59 or 3.588… or **SC1** for $\sqrt[32]{3.09}$ or 1.0358 to 1.036 seen
3(a)(i)	427 or 427.2 to 427.3…	**2**	**M1** for $\pi \times 8 \times 17$
3(a)(ii)	1010 or 1005….	**4**	**M2** for $\sqrt{17^2 - 8^2}$ oe or **M1** for $h^2 + 8^2 = 17^2$ oe **M1** for $\frac{1}{3} \times \pi \times 8^2 \times$ *their h* oe
3(a)(iii)	804 or 804.2 to 804.4 or 808	**1**	**FT** *their* **(ii)** × 0.8
3(a)(iv)	396 or 395.6 to 395.8 or 392	**1**	**FT** 1200 – *their* **(iii)**
3(b)(i)	$\frac{1}{54}$	**4**	**B3** for $\frac{\frac{4}{3}\pi r^3}{72\pi r^3}$ or better or **M2** for $\frac{\frac{4}{3} \times \pi \times r^3}{\pi \times (3r)^2 \times 8r}$ or $72 \times \pi \times r^3$ or **M1** for $\pi \times (3r)^2 \times 8r$ If 0 scored, **SC2** for answer of $\frac{1}{18}$
3(b)(ii)	972π final answer	**4**	**B2** for $r = \frac{9}{2}$ oe or **M1** for $4\pi r^2 = 81\pi$ or better **M1** for $2 \times \pi \times (3 \times$ *their r*$) \times (8 \times$ *their r*$)$ isw
4(a)	–1, 3	**2**	**B1** for each

Question	Answer	Marks	Partial Marks
4(b)	Correct graph	**3**	**B2FT** for 6 or 7 correct points or **B1FT** for 4 or 5 correct points
4(c)	Correct ruled tangent and $-2 \leqslant$ gradient $\leqslant -1.5$	**3**	**B2** for close attempt at tangent at $x = -4$ and answer in range OR **B1** for ruled tangent at $x = -4$ with no daylight and **M1** for rise/run **also dep on** close attempt at tangent. Must see correct or implied calculation from a drawn tangent.
4(d)	–3, 3	**1**	
4(e)	Correct graph	**4**	**B3FT** for 7 or 8 correct points or **B2FT** for 5 or 6 correct points or **B1FT** for 3 or 4 correct points
4(f)(i)	3.6 to 3.85	**1**	
4(f)(ii)	$x >$ *their* **(f)(i)**	**1**	**FT**
4(g)	$\frac{x^2}{4} = \frac{9}{x} + \frac{4}{x}$ or $\frac{x^3}{4} - 4 = 9$	**M1**	Allow $\frac{13}{x}$ for $\frac{9}{x} + \frac{4}{x}$
	52	**A1**	
5(a)(i)	265 or 265.3 to 265.4 nfww	**4**	**M1** for mid-values 150, 225, 275, 400 soi **M1** for Σfx where x is in correct interval including boundaries **M1 dep** for $\Sigma fx \div 52$ dependent on second **M1**
5(a)(ii)	Correct histogram	**4**	**B1** for each correct block If 0 scored, **SC1** for the four frequency densities seen
5(b)(i)	100	**1**	
5(b)(ii)	56	**1**	
5(b)(iii)	62	**1**	
5(b)(iv)	24	**1**	
5(b)(v)	88	**2**	**M1** for evidence of 12 written

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